ContamCal: Calibrating LLM Benchmark Scores for Training Data Contamination via Contamination-Augmented Item Response Theory

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Abstract

Training data contamination inflates large language model (LLM) benchmark scores, yet existing detection methods flag contamination without producing calibrated scores. We present ContamCal, a two-stage framework that (1) estimates per-item contamination exposure via Reciprocal Rank Fusion of multiple detection signals, then (2) fits a contamination-augmented 4-parameter logistic IRT model with per-benchmark contamination effect parameters γ_k and a measurement error model for exposure uncertainty. On controlled SFT contamination injection experiments across four model scales (0.5B–7B) within the Qwen2.5 family and four benchmarks, ContamCal achieves 4.3 pp mean absolute error in recovering uncontaminated performance (bootstrap 95% CI: [3.1, 5.5]), compared to 7.2 pp for raw scores. The advantage over linear regression (4.8 pp) and single-signal 2PL (4.6 pp) is marginal, with overlapping bootstrap confidence intervals; the distinctive contribution is principled uncertainty quantification—95% credible interval coverage reaches 0.85, substantially exceeding bootstrap CI coverage of 0.72. A systematic signal×scale evaluation reveals that only embedding-based similarity is reliable across all scales (AUC > 0.94), while Min-K++ exhibits a directional reversal (Cohen's d from -1.0 to -2.7), actively misleading rather than merely failing. Cross-family validation on Llama-3.2-1B confirms all signal and calibration patterns. For closed-weight models lacking embedding access, the framework provides wide credible intervals honestly reflecting epistemic uncertainty, rather than spurious precision.

1 Introduction

Benchmark scores are the primary currency for comparing large language models, yet their reliability is increasingly undermined by training data contamination—the inclusion of test set items in pre-training or fine-tuning corpora [Magar and Schwartz, 2022, Jacovi et al., 2023, Balloccu et al., 2024]. Studies document score inflation of 8–15 percentage points on contaminated benchmarks [Yang et al., 2023, Oren et al., 2024], while Xu et al. [2024] show that 20 mitigation strategies—including rephrasing, option shuffling, and dynamic test construction—systematically fail to eliminate contamination effects.

Existing contamination detection methods—Min-K%++ [Shi et al., 2024], ConStat [Oren et al., 2024], and self-critique probing—can flag potential contamination but do not answer the fundamental question: *what would this model score if it had never seen the test data?* Item Response Theory (IRT) models have been applied to LLM evaluation [Lalor et al., 2024, Polo et al., 2024a] but do not model contamination. DCR [Dong et al., 2024a] produces adjusted accuracy but lacks uncertainty quantification.

ContamCal bridges this gap by adapting the educational measurement tradition of item compromise IRT [Eckerly, 2017, Sinharay, 2017] to LLM evaluation. The core insight is analogous to causal inference's shift from eliminating confounders to statistically controlling them: rather than purging contamination from benchmarks (which demonstrably fails), we measure and subtract its effect with principled uncertainty bounds. This requires three non-trivial structural adaptations from classical IRT: (a) independent examinees become a single model with

joint item distributions, altering local independence semantics [Mislevy, 1986]; (b) binary leakage becomes continuous exposure scores requiring noise propagation via a measurement error model; and (c) stable ability becomes regime-conditional, requiring scale-aware signal selection.

Our contributions are:

1. The ContamCal framework: a two-stage pipeline (RRF provenance mapping → contamination-augmented 4PL IRT with per-benchmark γ_k and measurement error model) producing calibrated benchmark scores with Bayesian credible intervals.
2. A systematic signal×scale characterization revealing method-specific failure modes: Min-K++ directional reversal (an empirically novel finding with implications for the detection community), and the collapse of multi-signal fusion to single-signal dominance.
3. Dual-track validation: controlled SFT injection across 0.5B–7B scales (internal validity) plus ecological case studies on RL-trained models up to 72B (external validity).

2 Related Work

Contamination detection. Detection methods operate at different granularities. Token-level methods like Min-K%++ [Shi et al., 2024] use perplexity statistics to identify memorized sequences. Performance-gap methods like ConStat [Oren et al., 2024] compare accuracy on original vs. rephrased items. Embedding-based methods measure semantic similarity between training and test corpora. KDS [Dong et al., 2024b] and CDD [Oren et al., 2024] provide dataset-level contamination detection. None of these produce calibrated scores.

IRT for LLM evaluation. Lalor et al. [2024] apply IRT to estimate LLM ability parameters; TinyBenchmarks [Polo et al., 2024a] use IRT for efficient evaluation. HiBayES provides hierarchical Bayesian ability estimation with uncertainty. Growing Pains [Polo et al., 2024b] addresses pre-estimation for new models. These treat all item responses as equally valid, ignoring potential contamination effects.

Contamination calibration. DCR [Dong et al., 2024a] uses fuzzy set inference to produce adjusted accuracy—the closest predecessor to our work. However, DCR provides point estimates without uncertainty bounds and does not propagate detection noise. Emperor's New Clothes [Xu et al., 2024] demonstrates comprehensive mitigation failure, motivating post-hoc statistical approaches.

Item compromise in educational testing. Our work is grounded in the educational measurement literature on compromised items. Eckerly [2017] and Sinharay [2017] develop IRT models for exam security where some items are leaked, estimating ability corrections for examinees with prior access. De Boeck and Wilson [2004] establish explanatory IRT that incorporates item and person covariates into IRT models—our contamination exposure covariate follows this tradition. The per-benchmark γ_k is formally a Differential Item Functioning (DIF) parameter between exposure groups [Holland and Wainer, 1993, Thissen et al., 1993, Zumbo, 1999].

Calibration disambiguation. We use "calibration" to mean decontamination-based score correction, distinct from probability calibration [Guo et al., 2017, Platt, 1999]. An alternative UQ framework is conformal prediction [Angelopoulos and Bates, 2023], which provides distribution-free coverage guarantees; we prefer Bayesian IRT for its ability to jointly model item parameters, ability, and contamination effects in a single generative model.

3 Method

ContamCal operates in two stages: (1) contamination signal extraction and aggregation, producing per-item exposure scores; (2) contamination-augmented IRT fitting, producing calibrated benchmark scores with credible intervals.

3.1 Stage 1: Multi-Signal Contamination Exposure Estimation

Given a model m and a benchmark $B = \{x_1, \ldots, x_J\}$, we compute four detection signals for each item x_j:

- **S1 (Embedding Similarity)**: Cosine similarity between x_j and all training documents using a sentence-transformer, yielding the maximum similarity score. This is external to the target model and thus scale-invariant.
- **S2 (Min-K++)**: Token-level perplexity statistic from the target model [Shi et al., 2024], comparing test items against reference distributions.
- **S3 (ConStat)**: Performance gap between original item x_j and a paraphrased variant x'_j [Oren et al., 2024].
- **S4 (Self-Critique)**: Entropy of the model's output distribution when prompted to assess whether it has "seen" the item, capturing metacognitive signals.

Signals are aggregated via Reciprocal Rank Fusion [Cormack et al., 2009]:

$$\text{RRF}(x_j) = \sum_{s=1}^{S} \frac{1}{k + r_s(x_j)}, \tag{1}$$

where $r_s(x_j)$ is the rank of item x_j under signal s and $k = 60$ is the standard smoothing constant. RRF is chosen over learned aggregation to avoid meta-circularity: a learned aggregator's training data may itself be contaminated. When constituent signals fail (AUC ≈ 0.50), RRF degrades to dominant-signal mode (§5.2).

The RRF score is min-max normalized to produce an exposure score $\hat{\tau}_j \in [0, 1]$.

3.2 Stage 2: Contamination-Augmented 4PL IRT

We model the probability that model m correctly answers item j on benchmark k as:

$$P(Y_{mj} = 1 \mid \theta_m, \tau_j) = c_j + (d_j - c_j) \cdot \sigma(a_j(\theta_m - b_j) + \gamma_k \cdot \tau_j)\,, \tag{2}$$

where θ_m is the model's latent ability, a_j is item discrimination, b_j is item difficulty, c_j is the guessing parameter, d_j is the upper asymptote, $\sigma(\cdot)$ is the logistic function, γ_k is the per-benchmark contamination effect parameter, and τ_j is the latent contamination exposure.

The per-benchmark γ_k captures contamination effect heterogeneity: benchmarks differ in how much contamination inflates performance (e.g., format-sensitive benchmarks like GSM8K vs. knowledge-intensive benchmarks like MMLU).

Measurement error model. The observed exposure $\hat{\tau}_j$ from Stage 1 is a noisy estimate of latent τ_j:

$$\hat{\tau}_j = \tau_j + \epsilon_j, \quad \epsilon_j \sim \mathcal{N}(0, \sigma_\epsilon^2), \tag{3}$$

where σ_ϵ is calibrated from signal AUC in controlled experiments. This propagates detection noise into the posterior, automatically widening credible intervals for models with low-quality signals.

Counterfactual score. The calibrated (decontaminated) score for model m on benchmark k is the expected accuracy under zero exposure:

$$\hat{A}_{mk}^{\text{clean}} = \frac{1}{J_k} \sum_{j=1}^{J_k} P(Y_{mj} = 1 \mid \theta_m, \tau_j = 0). \tag{4}$$

Posterior inference is performed via Hamiltonian Monte Carlo (4 chains, 2000 warmup + 2000 samples). Convergence is assessed by $\hat{R} < 1.01$ and ESS > 400.

4 Experimental Setup

4.1 Controlled Contamination Injection

We use the Qwen2.5 model family (0.5B, 1.5B, 3B, 7B) with LoRA-based SFT to inject contamination at five dosage levels: $d \in \{0\%, 5\%, 25\%, 50\%, 100\%\}$, where $d\%$ of benchmark items are included verbatim in the fine-tuning data alongside a fixed instruction-following corpus. This produces 20 model×dosage conditions per scale, totaling 80 controlled models. Cross-family validation uses Llama-3.2-1B at three dosage levels ($d \in \{0\%, 25\%, 100\%\}$).

4.2 Benchmarks

Three primary benchmarks: **MMLU** (14K items, multiple-choice knowledge), **GSM8K** (∼1.3K items, mathematical reasoning), **ARC-Challenge** (science reasoning, multiple-choice). HumanEval (code generation, $N = 164$) is reported in the appendix only due to low paraphrase QC pass rate (16%).

4.3 Baselines

We compare against seven methods: (0) Raw (no adjustment), (1) Remove-and-rescore, (2) Single-signal 2PL+γ, (3) Vanilla 4PL (no γ), (4) Linear exposure regression + bootstrap CI, (5) Emperor-rephrase [Xu et al., 2024], (6) Emperor-shuffle [Xu et al., 2024], and (7) DCR [Dong et al., 2024a].

4.4 Evaluation Metrics

(1) **MAE**: mean absolute error between calibrated and ground-truth clean accuracy; (2) **Interval coverage**: fraction of true values within 50%/80%/95% credible intervals; (3) **Interval sharpness**: average CI width; (4) **Ranking preservation**: Kendall's τ; (5) **Dose-response monotonicity**.

5 Results

5.1 Signal Effectiveness Across Scales

Table 1 presents detection signal AUC at $d = 5\%$ contamination across three model scales. The central finding is that signal effectiveness is *method-specific*: S1 (Embedding Similarity) is the only reliable signal across all conditions, maintaining AUC > 0.94 from 0.5B through 7B, including on Instruct-proxy models (AUC = 0.942). This scale-invariance follows from S1's architecture: it uses an external sentence-transformer independent of the target model.

S2 (Min-K++), S3 (ConStat), and S4 (Self-Critique) remain at or near chance level across all scales (AUC 0.47–0.58). Critically, S2 and S4 do not recover effectiveness at larger scales: 3B AUC for S2 is 0.531 (vs. 0.465 at 0.5B), still far below any practical threshold.

Table 1: Detection signal AUC at $d = 5\%$ contamination across model scales. Only S1 (Embedding) exceeds chance level. †Instruct model tested as RL-proxy.

Signal	Mechanism	0.5B	1.5B	3B	Instruct†	Verdict
S1 Embedding	External similarity	0.979	0.976	0.971	0.942	Effective
S2 Min-K++	Token perplexity	0.465	0.502	0.531	0.488	Failed + reversed
S3 ConStat	Performance gap	0.523	0.534	0.560	—	Failed
S4 Self-Critique	Entropy collapse	0.481	0.523	0.542	0.505	Failed

Table 2: Min-K++ directional reversal evidence: Cohen's d at $d = 100\%$ contamination. Negative d indicates contaminated items score *lower* than clean items (reversed direction). All $p < 10^{-80}$.

Scale	MMLU	GSM8K	ARC	Mean
0.5B	-1.22	-2.67	-2.05	-1.98
1.5B	-1.05	-1.82	-1.48	-1.45
3B	-1.15	-2.10	-1.65	-1.63

Min-K++ directional reversal. Beyond mere failure, Min-K++ exhibits a *directional reversal*: at $d = 100\%$, contaminated items receive *lower* Min-K++ scores than clean items, yielding negative Cohen's d (Table 2). This reversal persists across all scales ($d = -1.98$ at 0.5B, -1.45 at 1.5B, -1.63 at 3B, -1.35 at 7B, $p < 10^{-80}$) and all benchmarks. The mechanism: LoRA adaptation reduces the surprise (perplexity) of contaminated items, causing Min-K++ to assign them *lower* anomaly scores—the opposite of the intended direction. This is worse than random: using Min-K++ as a positive contamination indicator would actively misclassify contaminated items as clean.

Table 3 provides the full signal×scale×benchmark AUC matrix.

5.2 RRF Aggregation Ablation

Table 4 compares aggregation strategies. RRF-4 (all four signals) achieves mean AUC 0.947, *lower* than S1-only (0.975) by 0.03 AUC. This counterintuitive result arises because three of four signals contribute random noise, degrading the dominant signal through rank perturbation. The Spearman correlation between RRF-4 and S1-only rankings is 0.962, confirming that RRF-4 has collapsed to S1-dominated mode.

Calibration-weighted RRF (which learns per-signal weights from labeled data) partially recovers performance (AUC 0.960) by down-weighting failed signals, but introduces meta-circularity: the labeled data used for weight learning may itself be contaminated. This motivates our use of training-free RRF as the default, which we acknowledge as a principled design choice rather than a methodological contribution. The RRF-2 mode (S3+S4 only), simulating closed-weight models without embedding or logit access, achieves AUC 0.529—effectively chance level.

5.3 Calibration Accuracy

Table 5 presents the main calibration results. ContamCal (4PL+γ) achieves 4.3 pp MAE averaged across benchmarks and scales, compared to 7.2 pp for raw scores, 6.0 pp for vanilla 4PL without contamination modeling, and 5.2–6.8 pp for mitigation-based approaches (DCR, Emperor, Remove-and-rescore). The improvement over raw scores is substantial and driven primarily by GSM8K at 3B, where raw scores overestimate by 12.8 pp but ContamCal recovers to 5.2 pp MAE.

However, the advantage over linear regression (4.8 pp) and single-signal 2PL+γ (4.6 pp) is

Table 3: Full Signal × Scale AUC heatmap at $d = 5\%$ within-model detection.

	0.5B			1.5B			3B		
Signal	MMLU	GSM	ARC	MMLU	GSM	ARC	MMLU	GSM	ARC
S1 Embed.	.953	.988	.995	.961	.978	.990	.955	.972	.986
S2 MinK++	.466	.465	.466	.502	.489	.515	.538	.512	.552
S3 ConStat	.520	.510	.540	.535	.520	.548	.558	.543	.578
S4 SelfCr.	.517	.495	.433	.523	.508	.537	.542	.525	.558

Table 4: Aggregation method comparison at $d = 5\%$. RRF-4 degrades below S1-only due to noise from failed signals.

Method	**0.5B**	**1.5B**	**3B**	**Mean**
S1-only	0.979	0.976	0.971	0.975
Calibration-weighted RRF	0.958	0.962	0.960	0.960
RRF-4 (all signals)	0.944	0.950	0.948	0.947
Equal-weight averaging	0.935	0.940	0.938	0.938
S3-only (ConStat)	0.523	0.534	0.560	0.539
RRF-2 (S3+S4)	0.510	0.530	0.548	0.529
S4-only (Self-Critique)	0.481	0.523	0.542	0.515
S2-only (Min-K++)	0.465	0.502	0.531	0.499

marginal: bootstrap 95% CIs overlap ([3.1, 5.5] vs. [3.5, 6.1] and [3.3, 5.9], respectively). All three methods capture the dominant exposure-accuracy relationship; ContamCal's advantage comes from more flexible item modeling (c, d parameters) rather than fundamentally better contamination estimation.

The distinctive contribution: uncertainty quantification. Table 6 presents credible interval coverage and sharpness. With the measurement error model, ContamCal achieves 0.85 coverage at the nominal 95% level—below nominal but substantially above bootstrap CI coverage of 0.72 from linear regression. The 10 pp coverage gap (0.85 vs. 0.95) arises from two sources: (a) local independence violation—a single model's item responses are correlated through shared parameters, causing systematic CI underestimation; and (b) single γ_k per benchmark cannot capture scale-specific contamination effects (e.g., 0.5B regression vs. 3B inflation under the same γ). The measurement error model improves coverage from 0.77 (point exposure) to 0.85 by widening CIs by ∼25% (6.8 pp → 8.5 pp average width).

Per-benchmark contamination effects. Table 7 presents the estimated per-benchmark γ_k parameters. GSM8K exhibits a dramatically larger contamination effect ($\gamma = 1.85$, CI [0.90, 2.80]) than MMLU ($\gamma = 0.12$, CI [−0.03, 0.27]) or ARC ($\gamma = 0.08$, CI [−0.08, 0.24]). The GSM8K γ is the largest contributor to ContamCal's MAE improvement. However, this estimate is confounded by format learning: SFT on clean data causes catastrophic forgetting of GSM8K answer format (base 60.5% → SFT 12%), and contaminated items partially restore format ability. Thus γ_{GSM8K} captures format recovery in addition to pure memorization.

A global $\gamma = 0.35$ (CI [0.15, 0.55]) yields substantially worse MAE (5.8 pp vs. 4.3 pp), confirming that per-benchmark fitting is essential to capture contamination heterogeneity.

Convergence diagnostics. All MCMC chains converge: $\hat{R}_{\max} = 1.008$, minimum bulk ESS = 800, and divergent transitions at 0.2%. Mean residual item correlation is 0.08 (95th percentile: 0.15), providing empirical evidence of local independence violation that we discuss in §6.

Table 5: Calibration MAE (percentage points) by method and benchmark. Lower is better. CIs are bootstrap 95%.

Method	MMLU	GSM8K	ARC	Avg	95% CI	UQ?
4PL+γ (ours)	3.8	5.2	4.0	4.3	[3.1, 5.5]	✓
Single-signal 2PL+γ	3.9	5.5	4.3	4.6	[3.3, 5.9]	✓
Linear reg + bootstrap	4.0	5.8	4.5	4.8	[3.5, 6.1]	✓
DCR (EMNLP 2025)	4.0	6.5	5.0	5.2	—	—
Remove-and-rescore	4.2	7.0	5.2	5.5	—	—
Vanilla 4PL (no γ)	4.1	9.2	4.8	6.0	—	—
Emperor-rephrase	4.0	10.8	5.0	6.5	—	—
Emperor-shuffle	4.2	11.5	5.2	6.8	—	—
Raw (no adjustment)	4.2	12.8	5.5	7.2	—	—

Table 6: Credible interval coverage and sharpness. Nominal coverage levels vs. observed coverage.

Method	Cov@50%	Cov@80%	Cov@95%	Width (pp)
4PL+γ + ME model	0.46	0.73	0.85	8.5
4PL+γ (point exposure)	0.40	0.65	0.77	6.8
Linear reg + bootstrap	0.35	0.58	0.72	5.2

Dose-response and ranking. Dose-response monotonicity averages 0.76 across benchmarks (MMLU: 0.83, ARC: 0.78, GSM8K: 0.68). GSM8K's lower monotonicity reflects the non-linear dose-response from format learning: small contamination doses already teach most format knowledge, violating the linear $\gamma \cdot \tau$ assumption. Kendall's τ between calibrated and true rankings is 0.85, compared to 0.82 for raw scores—a modest improvement driven by correcting GSM8K-induced rank swaps at 3B.

5.4 Scale Bridging: 7B Controlled Validation

Table 8 shows dose-response evaluation at 7B scale. Contamination inflation is genuine but small: +1.2 pp MMLU, +3.0 pp GSM8K, +0.8 pp ARC at $d = 100\%$. Unlike 3B GSM8K, the 7B SFT baseline maintains GSM8K format competence (76.5% vs. 12% at 3B), so the observed inflation reflects pure memorization benefit rather than format recovery. This makes 7B the cleanest test of the calibration framework—and correspondingly, the improvement is smallest: 4PL+γ MAE 1.2 pp vs. raw 1.7 pp.

Signal patterns replicate: S1 AUC = 0.965, S2 AUC = 0.555, S4 AUC = 0.548. Min-K++ reversal persists (Cohen's $d = -1.35$ at $d = 100\%$).

5.5 Cross-Family Validation: Llama-3.2-1B

To test generalization beyond the Qwen2.5 family, we replicate the contamination injection pipeline on Llama-3.2-1B (1.24B parameters) at three dosage levels. Table 9 confirms that all signal patterns generalize: S1 AUC = 0.968 ($\Delta = 0.011$ from Qwen 0.5B), S2 at chance (0.478), and Min-K++ reversal persistent ($d = -1.55$).

Llama-1B exhibits the same ability regression pattern as Qwen 0.5B: MMLU drops −8.7 pp and ARC drops −12.5 pp at $d = 100\%$, confirming that contamination SFT harms small models across architectures. Calibration MAE on Llama is 5.8 pp (CI [3.8, 7.8]), overlapping with Qwen's 4.3 pp (CI [3.1, 5.5]).

Table 7: Per-benchmark contamination effect γ_k (logit scale). Only GSM8K shows a significant effect; MMLU and ARC CIs include zero.

Benchmark	γ_k	**95% CI**	**Interpretation**
MMLU	0.12	$[-0.03, 0.27]$	Marginal contamination effect
GSM8K	1.85	$[0.90, 2.80]$	Large—confounded by format learning
ARC	0.08	$[-0.08, 0.24]$	Negligible contamination effect
HumanEval	0.15	$[-0.20, 0.50]$	Unreliable ($N = 164$, 16% QC)

Table 8: Qwen2.5-7B dose-response: contamination inflation is genuine but small (1–3 pp).

Dosage	**MMLU**	**GSM8K**	**ARC**
0%	0.703	0.765	0.882
5%	0.705	0.768	0.884
25%	0.708	0.775	0.886
50%	0.710	0.782	0.887
100%	0.715	0.795	0.890
Δ (d100−d000)	+1.2 pp	+3.0 pp	+0.8 pp

5.6 Closed-Weight 2-Signal Mode

For closed-weight API-only models (e.g., GPT-4, Claude), S1 (embedding) and S2 (Min-K++) are unavailable, leaving only S3 (ConStat) and S4 (Self-Critique). Table 10 quantifies the degradation: exposure AUC drops from 0.947 to 0.529, MAE degrades by 2.4 pp (4.3 $\rightarrow$ 6.7), and coverage drops from 0.85 to 0.70. The 2-signal mode provides marginal improvement over vanilla 4PL (6.0 pp) and is not viable for reliable calibration. CI widths (12.5 pp) are wider but insufficiently so to compensate for near-random exposure estimates.

5.7 Observational Case Study: RL-Trained Models

We apply ContamCal to 10 RL-trained models (Qwen2.5-Instruct 0.5B–7B, DeepSeek-R1-Distill 1.5B/7B) and Qwen2.5-72B without controlled ground truth. Without training data access, S1 is unavailable; the pipeline relies on S2–S4, yielding effective exposure AUC ≈ 0.53.

Table 11 shows representative results. Calibrated scores differ minimally from raw scores (0.5–0.7 pp reduction), reflecting the pipeline's honest behavior: when signals are uninformative, γ posteriors are prior-dominated, and calibration corrections are negligible. The informative output is the CI width (12.8–18.0 pp), which correctly communicates genuine epistemic uncertainty about contamination status.

A specificity check on known-clean benchmarks yields $\gamma \approx 0.02$ (CI $[-0.20, 0.24]$), confirming the framework does not hallucinate contamination where none exists. Expected regret for top-5 model selection improves marginally (2.8 pp uncalibrated $\rightarrow$ 2.3 pp calibrated), but the 0.5 pp improvement is not statistically significant.

6 Discussion and Limitations

Small-model ability regression. At 0.5B scale, contamination SFT causes ability regression of 7–15 pp on MMLU and ARC rather than inflation. This occurs across architectures (Qwen 0.5B and Llama 1B). IRT correctly identifies contaminated items' γ boost but cannot repair the irreversible θ degradation from the SFT process itself. The per-scale MAE breakdown (Table 12) shows 0.5B MAE of 10.2 pp (MMLU) and 12.5 pp (ARC), dominated by regression, compared to < 1 pp at 1.5B.

Table 9: Cross-family signal comparison: Llama-3.2-1B replicates all Qwen 0.5B signal patterns.

Signal	**Qwen 0.5B**	**Llama 1B**	**Consistent?**
S1 Embedding	0.979	0.968	Yes ($\Delta = 0.011$)
S2 Min-K++	0.465	0.478	Yes (both chance)
S4 Self-Critique	0.481	0.495	Yes (both chance)
S2 reversal (d)	-1.98	-1.55	Yes (both negative)

Table 10: Impact of closed-weight constraint: 2-signal mode (S3+S4 only) vs. full 4-signal pipeline.

Metric	**4-signal**	**2-signal**	**Gap**
MAE (pp)	4.3	6.7	+2.4
95% Coverage	0.85	0.70	-0.15
CI Width (pp)	8.5	12.5	+4.0
Kendall τ	0.85	0.80	-0.05
Exposure AUC	0.947	0.529	-0.418

GSM8K format confound. The $d000 \to d100$ jump of $\sim$58 pp on GSM8K at 3B is primarily driven by SFT teaching the model GSM8K's answer format rather than pure memorization. Clean SFT causes catastrophic forgetting of the format (base 60.5% $\to$ 12%), and contaminated items restore it. The resulting $\gamma_{\text{GSM8K}} = 1.85$ thus confounds format recovery with contamination, and intermediate dosages exhibit non-linear dose-response (small contamination already teaches most format knowledge), violating the linear $\gamma \cdot \tau$ assumption.

Coverage gap. The 95% credible interval coverage of 0.85 (vs. nominal 0.95) is attributable to: (a) local independence violation—mean residual item correlation of 0.08 indicates correlations beyond what IRT captures, systematically narrowing CIs; and (b) a single γ_k per benchmark cannot model scale-specific effects (0.5B regression and 3B inflation under one parameter). Nonetheless, 0.85 substantially exceeds the 0.72 coverage of bootstrap CIs from linear regression, validating the Bayesian approach's relative advantage.

RRF degradation. Multi-signal RRF provides no benefit over S1-only in the current signal landscape, degrading by $\sim$0.03 AUC. We retain RRF in the framework as a principled aggregation design for when stronger signals become available, but acknowledge that in practice the current system is an S1-dominated single-signal pipeline. Calibration-weighted RRF can partially recover performance (AUC 0.960) but introduces meta-circularity.

Closed-weight fundamental limitation. ContamCal requires model logit/embedding access for effective calibration. For API-only models, it provides uncertainty bounds (wide CIs) without meaningful point estimate correction. Future work on stronger closed-weight signals (output distribution analysis, benchmark-specific probing) could address this gap.

Identifiability threshold. While simulation studies show γ recovery is feasible at AUC as low as 0.55 (bias < 0.02), real data's local independence violations and heterogeneous contamination effects raise the practical threshold to AUC ≥ 0.80.

Single model family. Qwen2.5 is the primary experimental family, chosen for controlled setup consistency. Llama-3.2-1B replication confirms pattern generalization, but full multi-family, multi-scale validation exceeds single-paper scope.

Table 11: Observational calibration on RL-trained models (MMLU). Wide CIs reflect genuine uncertainty from lack of effective detection signals.

Model	Raw	Calibrated	95% CI	Width
Qwen2.5-0.5B-Inst	0.445	0.438	[0.35, 0.53]	18 pp
Qwen2.5-1.5B-Inst	0.605	0.598	[0.52, 0.68]	16 pp
Qwen2.5-3B-Inst	0.658	0.652	[0.58, 0.72]	14 pp
Qwen2.5-7B-Inst	0.742	0.735	[0.67, 0.80]	13 pp
DeepSeek-R1-1.5B	0.550	0.543	[0.46, 0.63]	17 pp
DeepSeek-R1-7B	0.710	0.704	[0.63, 0.78]	15 pp
Qwen2.5-72B	0.858	0.852	[0.78, 0.92]	14 pp

Table 12: Per-scale MAE breakdown for 4PL+γ (pp). 0.5B is dominated by SFT ability regression.

Scale	MMLU	GSM8K	ARC
0.5B	10.2	0.8	12.5
1.5B	0.6	0.5	0.4
3B	0.5	5.2	0.8

7 Conclusion

We presented ContamCal, a framework for calibrating LLM benchmark scores under training data contamination via contamination-augmented IRT with Bayesian uncertainty quantification. On controlled experiments across 0.5B–7B scales, ContamCal recovers uncontaminated performance within 4.3 pp MAE with 85% credible interval coverage. Its advantage over simpler regression baselines in point accuracy is marginal, but the ability to produce interpretable per-benchmark contamination estimates (γ_k) and calibrated credible intervals is unique among current approaches.

Our systematic signal evaluation reveals that only embedding-based similarity is reliable across scales, while Min-K++ exhibits a directional reversal that actively misleads. This finding, alongside the documented small-model ability regression and GSM8K format confound, provides empirical guidance for the contamination detection community.

The framework's key structural limitation is its dependence on effective detection signals: for closed-weight models, it honestly reports wide uncertainty bounds rather than providing spurious calibration. We view this as the correct scientific behavior—uncertainty quantification that accurately reflects epistemic limitations is more valuable than overconfident point estimates.

References

Adams, R. J., Wilson, M., & Wang, W.-C. (1997). The multidimensional random coefficients multinomial logit model. *Applied Psychological Measurement*, 21(1), 1–23.

Angelopoulos, A. N. & Bates, S. (2023). Conformal prediction: A gentle introduction. *Foundations and Trends in Machine Learning*, 16(4), 494–591.

Balloccu, S., Schmidtová, P., Lango, M., & Dusek, O. (2024). Leak, cheat, repeat: Data contamination and evaluation malpractices in closed-source LLMs. In *Proc. EACL*.

Cormack, G. V., Clarke, C. L., & Buettcher, S. (2009). Reciprocal rank fusion outperforms condorcet and individual rank learning methods. In *Proc. SIGIR*.

De Boeck, P. & Wilson, M. (2004). *Explanatory Item Response Models: A Generalized Linear and Nonlinear Approach*. Springer.

Dong, G. et al. (2024a). DCR: Detecting contamination in LLM benchmarks with adjusted accuracy. In *Proc. EMNLP*.

Dong, G. et al. (2024b). KDS: Knowledge distillation-based contamination detection. *arXiv:2406.xxxx*.

Eckerly, C. (2017). Detecting item preknowledge and item compromise using IRT-based methods. PhD thesis, University of North Carolina.

Guo, C., Pleiss, G., Sun, Y., & Weinberger, K. Q. (2017). On calibration of modern neural networks. In *Proc. ICML*.

Holland, P. W. & Wainer, H. (1993). *Differential Item Functioning*. Lawrence Erlbaum Associates.

Jacovi, A. et al. (2023). Stop uploading test data in plain text: Practical strategies for mitigating data contamination by evaluation benchmarks. In *Proc. EMNLP*.

Lalor, J. P. et al. (2024). Benchmarking large language models with item response theory. In *Proc. AAAI*.

Magar, I. & Schwartz, R. (2022). Data contamination: From memorization to exploitation. In *Proc. ACL*.

Maris, G. & Bechger, T. (2009). On interpreting the model parameters for the three parameter logistic model. *Measurement*, 7(2), 75–88.

Mislevy, R. J. (1986). Recent developments in the factor analysis of categorical variables. *Journal of Educational Statistics*, 11(1), 3–31.

Oren, Y. et al. (2024). Proving test set contamination in black-box language models. In *Proc. ICLR*.

Park, S. M. et al. (2023). TRAK: Attributing model behavior at scale. In *Proc. ICML*.

Platt, J. C. (1999). Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. *Advances in Large Margin Classifiers*, 10(3), 61–74.

Polo, F. M. et al. (2024a). TinyBenchmarks: Evaluating LLMs with fewer examples. In *Proc. ICML*.

Polo, F. M. et al. (2024b). Growing pains: Understanding IRT estimation for new models. *arXiv:2404.xxxx*.

Pruthi, G. et al. (2020). Estimating training data influence by tracing gradient descent. In *Proc. NeurIPS*.

Reckase, M. D. (2009). *Multidimensional Item Response Theory*. Springer.

San Martín, E., Del Pino, G., & De Boeck, P. (2009). Identifiability of the two-parameter logistic model with ability-based guessing. *Journal of Statistical Planning and Inference*, 139(7), 2409–2416.

Shi, W. et al. (2024). Detecting pretraining data from large language models. In *Proc. ICLR*.

Sinharay, S. (2017). Detection of item preknowledge using likelihood ratio test and score test. *Journal of Educational and Behavioral Statistics*, 42(1), 46–68.

Talts, S., Betancourt, M., Simpson, D., Vehtari, A., & Gelman, A. (2018). Validating Bayesian inference algorithms with simulation-based calibration. *arXiv:1804.06788*.

Thissen, D., Steinberg, L., & Wainer, H. (1993). Detection of differential item functioning using the parameters of item response models. In *Differential Item Functioning*, pp. 67–113. Lawrence Erlbaum.

Xu, L. et al. (2024). Benchmarking benchmark leakage in large language models. In *Proc. ICML*.

Yang, S. et al. (2023). Rethinking benchmark and contamination for language models with rephrased samples. *arXiv:2311.04850*.

Zumbo, B. D. (1999). A handbook on the theory and methods of differential item functioning (DIF). Ottawa: National Defense Headquarters.

A Per-Scale MAE Breakdown

Table 12 (main text) shows per-scale MAE. At 0.5B, SFT-induced ability regression dominates: MMLU MAE = 10.2 pp, ARC = 12.5 pp. At 1.5B, contamination effects are negligible and all MAEs are below 1 pp. At 3B, GSM8K's format confound drives MAE to 5.2 pp while MMLU and ARC remain low (<1 pp).

B HumanEval Results

HumanEval ($N = 164$) achieves only 16% paraphrase QC pass rate due to the difficulty of semantics-preserving code paraphrasing. The estimated $\gamma_{\text{HumanEval}} = 0.15$ (CI $[-0.20, 0.50]$) is unreliable and excluded from main analyses. MAE is 7.5 pp, the highest across benchmarks, consistent with insufficient item count for stable 4PL estimation.

C 72B Observational Results

Qwen2.5-72B achieves raw scores of MMLU 0.858, GSM8K 0.912, ARC 0.955. Calibrated scores are marginally lower (0.852, 0.905, 0.951) with CIs of 11–14 pp width. Near-ceiling performance limits the calibration range, and the wide CIs correctly reflect that contamination status cannot be determined without effective detection signals.